HALEY PATEL

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EDUCATION

New Jersey Institute of Technology, *Honor's Scholar* — Rising Junior, GPA: **3.92/4.0**

B.S. Computer Science Minor: Applied Mathematics, *Ying Wu College of Computing* — Expected Graduation: **May 2025**

- Received full honor's merit scholarship for all years of attendance.
- **Relevant Coursework:** Data Structures and Algorithms, Database Systems, Java, C/C++, Python, Statistics, Multivariable Calculus, Actuarial Mathematics, Programming in Linux (in progress), Discrete Mathematics, Quantitative Analysis

EXPERIENCE

Machine Learning for Advanced Manufacturing Optimization — ***May 2023 - Present***

Honor's Summer Research Institute Researcher

- Transformed **9 million data points** to produce an **iterative machine learning algorithm** that simulates and predicts temperature profiles of particles in a laser manufacturing process with **99.97%** accuracy.
- Utilized an Anaconda environment for the seamless use of Python NumPy, Pandas, Scikit-Learn, and PyTorch libraries.

Software Development Intern, *UPS (United Parcel Service)* — ***June 2023 – Present***

- Spearheading an **employee user experience research** project by gathering and **analyzing application usage data** to pinpoint inefficiencies or confusions in employee workflows to improve employee user experience with enterprise applications.
- Assigned to develop a full-stack employee email server management website to increase the efficiency of migrating employee email servers across regions. Created a **Jira** and **Confluence** board to track development progress. (C#, C++, Kubernetes)

Data and Claims Intern, *Plymouth Rock Assurance* — **June 2022 – August 2022**

- Utilized insurance department's data management systems to support data entry and analysis tasks, increasing personal efficiency in resolving claims by **40%** by the end of the internship.

PROJECTS

Banking Website and Database (MySQL, PHP, HTML5, CSS, Oracle SQL Server) — ***February – May 2023***

- Conceptualized and built a **database on Oracle SQL Server** to replicate the requirements for a fictional banking institution.
- Utilized **PHP** and **MySQL** to establish the **backend infrastructure** for the operational banking website. Designed with **HTML5** and **CSS**, the website includes login pages, customer transaction features, and an employee management dashboard.

Task Organization Application (RESTful API, PostgreSQL, Node.js, Express, React) — ***June 2023***

- Developed **full-stack, user-friendly task list application** by developing a **RESTful API** using **Node.js** and **Express** to enable seamless communication between the front-end and back-end.
- Utilized **PostgreSQL** to store and manage task data and built the front end with **React**, **Bootstrap, HTML5, JSX**.

Compiler for SPL Language (C++) — ***February – April 2023***

- Built a **working compiler** for a Simple Perl-Like Language that includes a **lexical analyzer** and **parser interpreter.**
- Compiler **parses and interprets code** written in the SPL language and **produces correct** program **outputs or error messages.**

RSA Encryption Algorithm (Python) — ***May 2023***

- Programmed modular arithmetic functions and Euler's Totient Function from scratch to implement RSA encryption**. Improved runtime and computational cost** of running the program with large inputs by **200%** by implementing bitwise shifting.

LEADERSHIP AND EXPERIENCES

Vice President, *Business Masterminds* — ***January 2022 - Present***

- Initiated an annual collaboration with campus technology organizations to host an **Artificial Intelligence in Business** workshop of **50+ attendees** to teach how to leverage artificial intelligence tools to solve common business problems.
- Maintained connections with **10+ industry professionals** and school alumni to serve as guest speakers in workshops/events.

Financial Analyst, *Investment Fund NJIT* — ***September 2021 - Present***

- Analyzed **financials** of publicly traded companies discovered through **industry/market research** to find viable, long-term growth investment opportunities to increase the value of the club's stock portfolio. (**Bloomberg Terminal, EDGAR**)

SKILLS

Technology: Python (PyTorch, Numpy, Scikit-Learn, Pandas), C, C++, C#, Java, OOP, JavaScript, HTML5, CSS, SQL, MySQL, PostgreSQL, PHP, Nodejs, Express, React, RESTful APIs, Jira, Tableau, 3D modeling, creative arts

Certifications: Microsoft Office Specialist Word 2016-2019, Microsoft Office Specialist PowerPoint 2016-2019